

MAE 3127

Fluid Mechanics Laboratory



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Topics covered today:
Hydrostatics and Gates



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& Applied Science

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Photo: Kartik Bulusu

Preliminaries

Center of Gravity

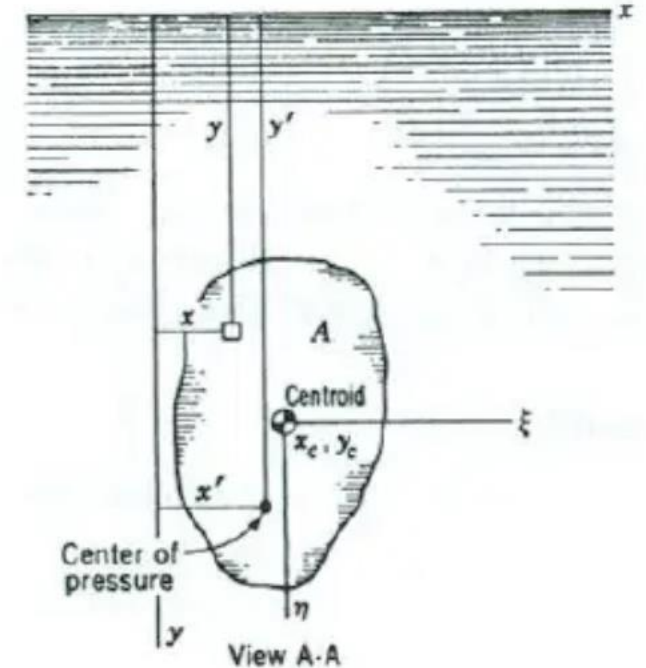
The **center of gravity** is the average location of the weight of an object.

- Describes the motion of any object through space in terms of the **translation** and/or **rotation**
- **Application:** If the **object is confined to rotate** about some other point, **like a hinge**, we can still describe its motion.

Center of Pressure

The **center of pressure** is the point on a body where the resultant force and center of pressure location produce an equivalent force and moment on the body as the original pressure field.

- **Alternative definition:** The point on a surface where the resultant hydrostatic force acts.
- **Application:** This **concept is used for understanding structures like dams and gates**, where the total force of the water needs to be determined for design and stability calculations.



Preliminaries

First Moment of Area

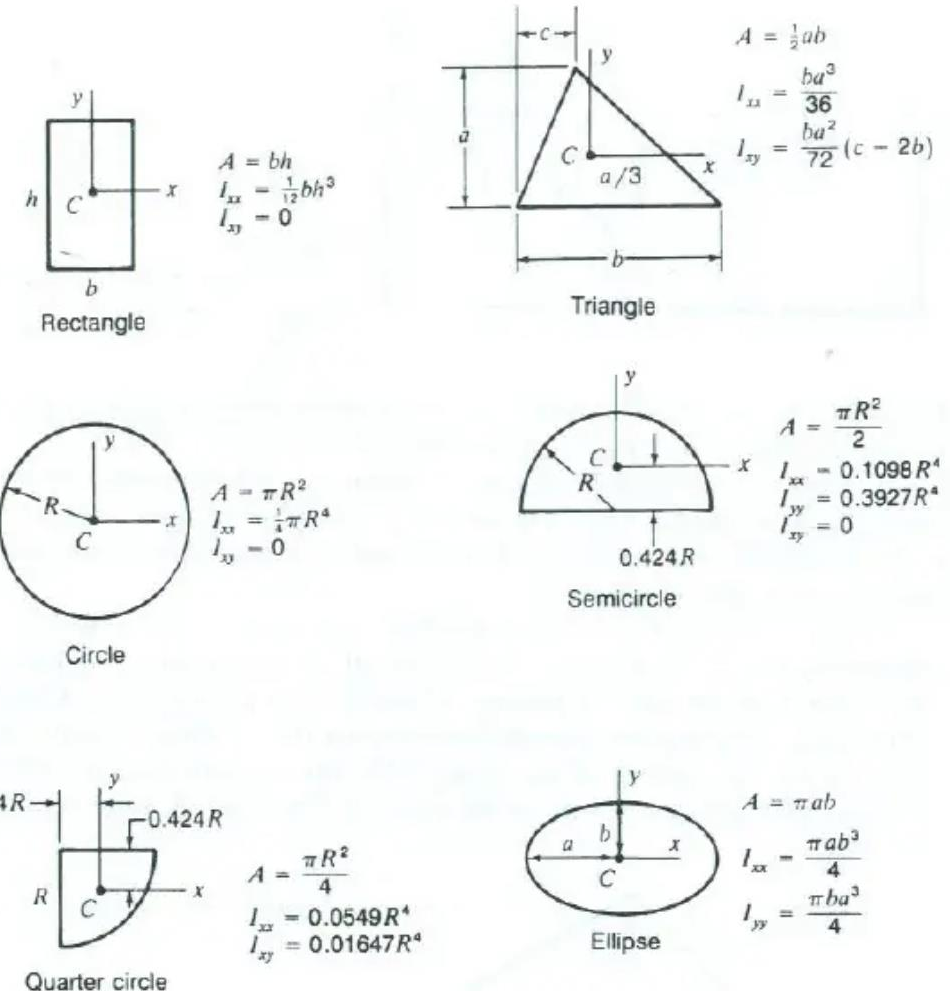
The first moment of area, or [statical moment](#), is a mathematical quantity that measures the distribution of an area's shape relative to a specific axis.

- The first moment of area of a shape, about a certain axis, equals the sum over all the [infinitesimal](#) parts of the shape of the area of that part times its distance from the axis.

Second Moment of Area

The second moment of area (also known as the moment of inertia) quantifies a shape's resistance to bending and deformation when a load is applied, indicating how efficiently the cross-sectional area resists bending.

- The primary physical meaning of the second moment of area is its role in resisting bending.
- A larger second moment of area means the material is "further away" from the bending axis, making the structure stiffer and more resistant to deformation



Given

- You are given a plane submerged in a liquid.
- Find the force acting on it and its location.

Strategy

Step 0: Understand the problem

- Identify the fluid and its properties
- Identify the geometry of the gate
- Look for the pressures given at the free surface

Step 1: Calculate the magnitude of the reaction force on the plate ($|F_R|$)

- Find the location of the center of gravity (h_c)
- Calculate the hydrostatic pressure at that location ($p_c = \gamma h_c$)
- Calculate the surface area of the plate (A)
- Calculate the magnitude of the reaction force ($|F_R| = \gamma h_c A$)

Step 2: Calculate the location of the reaction force on the plate (F_R)

- The location of F_R is the center of pressure ($y' - y_c$)
- Note that y_c is the inclined distance of the center of gravity. So, $y_c = \frac{h_c}{\sin(\theta)}$

$$y' - y_c = \frac{\gamma \sin \theta I_{\zeta\zeta}}{p_c A}$$

